

Independent Study Spring 2026 - 1D Dynamical Systems

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0 Introduction

This is a compilation of all my independent study work produced over the course of the Spring 2026 semester. This independent study covers one dimensional dynamical systems, exploring this with the following function and map:

$$f(x) = ax(1 - x) \text{ defined on } f : [0, 1] \rightarrow [0, 1]$$

Contents

0	Introduction	1
1	Weeks 1-2	1
1.1	Premise	1
1.2	Finding critical points	2
1.3	Finding values of a	2
1.4	Finding fixed points in terms of a	2
1.5	Finding attracting fixed points	3
2	Weeks 3-7	3
2.1	Premise	3
2.2	Finding values of a s.t. $S_f(x) < 0$	3
2.3	Finding $S_f(x)$ when $f(x) = \frac{ax+b}{cx+d}$	3
2.4	Finding values of a such that $(f^2(x)) < 1$ for periodic points of order 2	4
2.4.1	Finding critical points for the defined range of x	4
2.4.2	Finding values of a	5
2.4.3	Finding periodic points in terms of a	5
2.4.4	Finding attracting periodic points on order 2	6
2.5	Finding the first five terms of V	6
2.6	Discussion on section 2.5	7
3	Weeks 8-9	8
3.1	Premise	8
3.2	Python program to find first n terms of V on $f(x) = ax(1 - x)$	8
3.3	Parity-Lexicographical Ordering	8
3.4	Python program to find values of $ax(1 - x)$ given string of 0s and 1s	9
4	Week 10	10
4.1	Premise	10
4.2	Python program to find pre-image	10
4.3	All possible a for first three terms of V	10
5	Week 11	11
5.1	Premise	11
5.2	Python program to find pre-image of pre-image (and beyond)	11
5.3	All possible a for first four terms of V	13
5.4	Degree 3 polynomials with two critical points on map and single parameter	14

1 Weeks 1-2

1.1 Premise

This section details the finding of attracting fixed points on the function

$$f(x) = ax(1 - x)$$

where a is a parameter. This function is defined under the following:

$$f : [0, 1] \rightarrow [0, 1]$$

Finding these attracting fixed points involves four steps:

1. Finding critical points for the defined range of x
2. Finding values of $a \neq 0$ that make the function defined
3. Finding fixed points in terms of a
4. Using information from the previous steps to find values of a that have an attracting fixed point, both when $x = 0$ and when $x \neq 0$.

1.2 Finding critical points

Consider

$$f(x) = ax(1 - x)$$

where $0 \leq x \leq 1$.

Definition 1.1: A critical point is when $f'(x) = 0$ or when the function is undefined.

The function is defined for all possible x , so no need to worry about that.

First, find the derivative of $f'(x)$:

$$\begin{aligned} f(x) &= ax(1 - x) \\ f(x) &= ax - ax^2 \\ f'(x) &= a - 2ax \end{aligned}$$

Next, set the function to 0 and solve for x :

$$\begin{aligned} a - 2ax &= 0 \\ a &= 2ax \\ 1 &= 2x \\ \frac{1}{2} &= x \end{aligned}$$

Thus, the only critical point is $x = \frac{1}{2}$.

1.3 Finding values of a

Objective: Find $a \neq 0$ such that $0 \leq f(x) \leq 1$.

Observe that having $a < 0$ will result in $f(x) \leq 0$. $x(1 - x)$ is positive (zero at $f(0)$ and $f(1)$) for all possible x , so multiplying this by a negative term results in a negative number.

For $a > 0$, $f(x) \geq 0$ since multiplying $x(1 - x)$ by a positive number gives a positive number. Also note that $f'(x)$ is positive for all $a > 0$ for $0 \leq x < \frac{1}{2}$ and negative for $\frac{1}{2} < x \leq 1$. This implies that the critical point at $x = \frac{1}{2}$ is a relative maximum.

The value of the critical point strictly increases with a . Therefore, the maximum value that a can be is when $f(\frac{1}{2}) = 1$. This occurs when $a = 4$. As such, given $0 \leq x \leq 1$, $a \neq 0$:

$$0 < a \leq 4$$

1.4 Finding fixed points in terms of a

Definition 1.2: A fixed point is defined when $f(x) = x$

To find fixed points in terms of a , solve for x .

$$\begin{aligned} ax(1 - x) &= x \\ ax - ax^2 &= x \\ a(x - x^2) &= x \\ a &= \frac{x}{x - x^2} \\ a &= \frac{x}{x(1 - x)} \\ a &= \frac{1}{1 - x} \\ a^{-1} &= 1 - x \\ a^{-1} - 1 &= -x \\ 1 - \frac{1}{a} &= x \\ \frac{a-1}{a} &= x \end{aligned}$$

Every x that satisfies this condition is a fixed point. There is also a fixed point at $x = 0$ as whatever the value of a is, $f(0) = 0$.

1.5 Finding attracting fixed points

Definition 1.3: A fixed point is an attractor when $|f'(x)| < 1$, indifferent when $|f'(x)| = 1$, and repelling when $|f'(x)| > 1$.

When $x = 0$:

$$f'(x) = a - 2a(0) = a$$

The fixed point at $x = 0$ is therefore attracting when $a < 1$. (Note that a is still subject to the bounds defined in section 1.3 - "Finding values of a ".)

When $x \neq 0$:

Express the derivative in terms of a :

$$\begin{aligned} f'(a) &= a - 2a\left(\frac{a-1}{a}\right) \\ &= -a + 2 \end{aligned}$$

Given the constraints of a , the derivative is strictly decreasing, so there will be only one time when the function goes below 1 and then below -1 . When a is between these values, the fixed point will be attracting.

The function is 1 when $a = 1$ and is -1 when $a = 3$. Therefore, when $x \neq 0$, any value $1 < a < 3$ produces an attracting fixed point.

2 Weeks 3-7

2.1 Premise

This section builds off the previous section introduces Schwartzian derivatives, and covers some results from that. Next, it finds attracting periodic points on $f(f(x))$, which has a similar process of finding fixed points on $f(x)$. Finally, it explores finding the first five terms of V , which will be defined when it becomes relevant.

2.2 Finding values of a s.t. $S_f(x) < 0$

Remember from last time that $0 < a \leq 4$

Definition 2.1: The Schwartzian derivative is defined as

$$S_f(x) = \frac{f'''(x)}{f'(x)} - \frac{3}{2} \left(\frac{f''(x)}{f'(x)} \right)^2$$

Calculating the Schwartzian derivative for $f(x)$:

$$\begin{aligned} f(x) &= ax - ax^2 \\ f'(x) &= a - 2ax \\ f''(x) &= -2a \\ f'''(x) &= 0 \end{aligned}$$

Nothing needs to be plugged in to solve this besides $f'''(x)$. Observe that $f'''(x) = 0$, so the first term is completely canceled out. ($f''(x) \neq 0$) given the bounds of a , so the first term is always defined, it's just always 0. The second term is something squared. Anything squared is non-negative - because this value is being subtracted from 0, $S_f(x)$ cannot be larger than 0. Furthermore, the numerator is defined in terms of a , which is always nonzero, so $S_f(x)$ is always negative provided the function is defined.

Because $S_f(x)$ is always negative, **any value of a makes the Schwartzian derivative negative.**

2.3 Finding $S_f(x)$ when $f(x) = \frac{ax+b}{cx+d}$

Calculating the Schwartzian derivative for $f(x)$ (note: simplification steps not shown):

$$\begin{aligned} f(x) &= \frac{ax+b}{cx+d} \\ f'(x) &= \frac{ad-bc}{(cx+d)^2} \\ f''(x) &= \frac{-2c(ad-bc)}{(cx+d)^3} \\ f'''(x) &= \frac{6c^2(ad-bc)}{(cx+d)^4} \end{aligned}$$

So the Schwartzian derivative is

$$\frac{\frac{6c^2(ad-bc)}{(cx+d)^4}}{\frac{ad-bc}{(cx+d)^2}} - \frac{3}{2} \left(\frac{\frac{-2c(ad-bc)}{(cx+d)^3}}{\frac{ad-bc}{(cx+d)^2}} \right)$$

This simplifies into

$$\frac{6c^2}{(cx+d)^2} - \frac{3}{2} \left(\frac{-2c}{cx+d} \right)^2 = 0$$

This is defined when $cx+d \neq 0$ and $ad-bc \neq 0$.

2.4 Finding values of a such that $(f^2(x)) < 1$ for periodic points of order 2

This part is like the previous report, but is on $f(f(x))$ instead. There are a few differences outside this. Below is the general process involved in achieving the above goal.

1. Finding critical points for the defined range of x
2. Finding values of $a \neq 0$ that make the function defined
3. Finding periodic points in terms of a
4. Using information from the previous steps to find values of a such that $f^2(x)' < 1$ for periodic points of order 2

Before proceeding, find the value of $f(f(x))$ (some simplification steps cut):

$$\begin{aligned} f(f(x)) &= (a^2x(1-x)(1-(ax(1-x)))) \\ &= a^2x(1-x) - a^2x(1-x)(ax(1-x)) \\ &= a^2x(1-x) - a^3x^2 - 2a^3x^3 + a^3x^4 \\ &= a^2x(1-x) - a^3x^2(1-2x+x^2) \\ &= a^2x(1-x) - a^3x^2(x-1)^2 \end{aligned}$$

The end result is what will be used when referencing $f(f(x))$, or $f^2(x)$.

2.4.1 Finding critical points for the defined range of x

To find critical points, set the derivative equal to zero.

$$f^2(x)' = a^2(1-x) - a^2x - 2a^3x(x-1)^2 - 2a^3x^2(x-1) = 0$$

The following shows some algebra manipulation to solve for x (some simplification steps skipped):

$$\begin{aligned} a^2(1-x) - a^2x - 2a^3x(x-1)^2 - 2a^3x^2(x-1) &= 0 \\ a^2(1-x) - a^2x &= 2a^3x(x-1)^2 + 2a^3x^2(x-1) \\ a^2(1-2x) &= 2a^3(x(x-1)^2 + x^2(x-1)) \\ 1-2x &= 2a(x(x-1)^2 + x^2(x-1)) \\ 2a &= \frac{1-2x}{x(x-1)^2 + x^2(x-1)} \\ 2a &= \frac{1-2x}{2x^3 - 3x^2 + x} \\ 2a &= \frac{1-2x}{(1-2x)(-x^2+x)} \\ 2a &= \frac{1}{(-x^2+x)} \\ \frac{1}{2a} &= -x^2+x \\ -x^2+x - \frac{1}{2a} &= 0 \\ x &= \frac{-1 \pm \sqrt{1 - \frac{4}{2a}}}{-2} \end{aligned}$$

Observe that unlike the previous critical point found for order 1, these critical points depend on a . Furthermore, $\frac{4}{2a}$ is only defined when the fraction is no bigger than 1. Therefore, the critical points found above are only defined when $a \geq 2$.

Critical points found in order 1 are also valid in order 2. So, to clarify, the 3 critical points for $f^2(x)$ are

$$x = \frac{1}{2}, x = \frac{-1 \pm \sqrt{1 - \frac{4}{2a}}}{-2} (a \geq 2)$$

2.4.2 Finding values of a

Plug in the critical points into $f^2(x)'$. For $x = \frac{1}{2}$:

$$\begin{aligned} a^2 - 2a^2 \left(\frac{1}{2}\right) - 2a^3 \left(\frac{1}{2}\right) \left(\frac{1}{2} - 1\right)^2 - 2a^3 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2} - 1\right) \\ \frac{1}{2}a^2 - \frac{1}{4}a^2 - \frac{1}{16}a^3 \\ \frac{1}{4}a^2 - \frac{1}{16}a^3 \end{aligned}$$

To find points where the derivative changes, set the derivative equal to 0.

$$\begin{aligned} \frac{1}{4}a^2 - \frac{1}{16}a^3 &= 0 \\ \frac{1}{4}a^2 &= \frac{1}{16}a^3 \\ \frac{1}{4} &= \frac{1}{16}a \\ a &= 4 \end{aligned}$$

$a = 0$ too:

$$\begin{aligned} \frac{1}{4}(0)^2 - \frac{1}{16}(0)^3 &= 0 \\ 0 &= 0 \end{aligned}$$

By plugging in arbitrary values for $a < 0, 0 < a < 4, a > 4$, determine whether the derivative increases or decreases at these values.

For $a < 0$ (choosing -1):

$$\frac{1}{4}(-1)^2 - \frac{1}{16}(-1)^3 = \frac{5}{16} \implies \text{increasing}$$

For $0 < a < 4$ (choosing 1):

$$\frac{1}{4}(1)^2 - \frac{1}{16}(1)^3 = \frac{3}{16} \implies \text{increasing}$$

For $a > 4$ (choosing 5):

$$\frac{1}{4}(5)^2 - \frac{1}{16}(5)^3 = -\frac{25}{16} \implies \text{decreasing}$$

Plug in values for a in $f^2(x)$ for our critical points of the derivative with $x = \frac{1}{2}$:

$$0^2 \frac{1}{2} \left(1 - \frac{1}{2}\right) - 0^3 \frac{1}{2} \left(\frac{1}{2} - 1\right)^2 = 0$$

Because the derivative was increasing prior to $a = 0$, this implies that $f^2(x) < 0$ for $a < 0$. Therefore, it still holds that $a > 0$.

Now do the same but for $a = 4$:

$$4^2 \frac{1}{2} \left(1 - \frac{1}{2}\right) - 4^3 \frac{1}{2} \left(\frac{1}{2} - 1\right)^2 = 0$$

This implies that the derivative grows throughout $0 < a \leq 4$ throughout. With everything stated above, this implies that a is still defined on $0 < a \leq 4$.

$$\text{Now find values of } a \text{ for } x = \frac{-1 \pm \sqrt{1 - \frac{4}{2a}}}{-2}$$

Using a computer algebra system (derivativecalculator.net), a is defined for $0 < a \leq 4$ on the map. So overall, a is defined for $0 < a \leq 4$ because all critical points have this range.

2.4.3 Finding periodic points in terms of a

Definition 2.2: A periodic point is defined as

$$f^n(x) = x, \text{ where } f^i(x) \neq x \forall 1 \leq i \leq n$$

A fixed point is this, but $n = 1$

To find periodic points of order 2, solve for x . Note that the complications get very complex, so I utilized Gemini (AI LLM) to find these periodic points.

$$x = \frac{a + 1 \pm \sqrt{a^2 - 2a - 3}}{2a}, a \geq 3$$

These results were validated by plugging in this x into Desmos and seeing how the values matched up with the original equation. (Note that the two fixed points defined in the previous report still hold here, but because they exist when $f^1(x) = x$, they are not periodic points.)

2.4.4 Finding attracting periodic points on order 2

Definition 2.3: A periodic point p is attracting when

$$\left| \frac{d}{dx} (f^n(p)) \right| < 1$$

A periodic point p is repelling when

$$\left| \frac{d}{dx} (f^n(p)) \right| > 1$$

A periodic point p is indifferent when

$$\left| \frac{d}{dx} (f^n(p)) \right| = 1$$

Like before, plug the fixed points into $f^2(x)'$.

$$\text{For } x = \frac{a+1+\sqrt{a^2-2a-3}}{2a}:$$

Making the whole thing in terms of a :

$$\begin{aligned} f^2(a)' &= a^2 \left(1 - \frac{a+1+\sqrt{a^2-2a-3}}{2a} \right) - a^2 \left(\frac{a+1+\sqrt{a^2-2a-3}}{2a} \right) \\ &\quad - 2a^3 \left(\frac{a+1+\sqrt{a^2-2a-3}}{2a} \right) \left(\left(\frac{a+1+\sqrt{a^2-2a-3}}{2a} \right) - 1 \right)^2 \\ &\quad - 2a^3 \left(\frac{a+1+\sqrt{a^2-2a-3}}{2a} \right)^2 \left(\left(\frac{a+1+\sqrt{a^2-2a-3}}{2a} \right) - 1 \right) \end{aligned}$$

Using Gemini to help simplify this, obtain $-a^2 + 2a + 4$. This simplification was verified by matching the original equation to this one in Desmos.

The periodic point is attracting when $|a^2 - 2a + 4| < 1$. Note that values $a < 3$ cannot be considered because the periodic points are undefined at this value.

At $a = 3$, the derivative is $|-3^2 + 2(3) + 4| = 1$. By obtaining the second derivative, $-2a + 2$, this shows that the derivative is constantly decreasing. So when the derivative drops below -1 , it won't come back up. This happens when $a = 1 + \sqrt{6}$ (the derivative is also -1 when $a = 1 - \sqrt{6}$, but this is out of the initial bounds). Also note: $1 + \sqrt{6} \approx 3.4495$

In conclusion, $\frac{a+1+\sqrt{a^2-2a-3}}{2a}$ is an attracting periodic point when $3 < a < 1 + \sqrt{6}$

$$\text{For } x = \frac{a+1-\sqrt{a^2-2a-3}}{2a}:$$

$$\begin{aligned} f^2(a)' &= a^2 \left(1 - \frac{a+1-\sqrt{a^2-2a-3}}{2a} \right) - a^2 \left(\frac{a+1-\sqrt{a^2-2a-3}}{2a} \right) \\ &\quad - 2a^3 \left(\frac{a+1-\sqrt{a^2-2a-3}}{2a} \right) \left(\left(\frac{a+1-\sqrt{a^2-2a-3}}{2a} \right) - 1 \right)^2 \\ &\quad - 2a^3 \left(\frac{a+1-\sqrt{a^2-2a-3}}{2a} \right)^2 \left(\left(\frac{a+1-\sqrt{a^2-2a-3}}{2a} \right) - 1 \right) \end{aligned}$$

This simplifies again to $-a^2 + 2a + 4$, so like $\frac{a+1+\sqrt{a^2-2a-3}}{2a}$, $\frac{a+1-\sqrt{a^2-2a-3}}{2a}$ is an attracting periodic point when $3 < a < 1 + \sqrt{6}$

Some quick follow-up questions (I myself don't know the answer to these):

1. Notice that the lower bound for attracting periodic points for order 2 ($a = 3$) is when the fixed points stopped being attracting. For order 3 periodic points, will they become attracting when $a > 1 + \sqrt{6}$?
2. If so, when will it stop being attracting? What about order 4? Will it follow the same pattern?
3. If so, do the ranges of these periodic points converge to a specific value? If so, What does that value represent?

2.5 Finding the first five terms of V

Definition 2.4: Let $V_1 V_2 \dots V_n$ be defined as the following:

First, plug in $x = \frac{1}{2}$ and some a in for x in $f(x, a) = ax(1-x)$. Then assign V_1 on the following:

$$\begin{aligned} V_1 &= 0 \quad \text{if} \quad f\left(\frac{1}{2}, a\right) < \frac{1}{2} \\ V_1 &= 1 \quad \text{if} \quad f\left(\frac{1}{2}, a\right) > \frac{1}{2} \end{aligned}$$

The sequence ends if $f\left(\frac{1}{2}, a\right) = \frac{1}{2}$.

If $f\left(\frac{1}{2}, a\right) \neq \frac{1}{2}$, the result is the new value that's plugged in for x . V_2 is obtained the same way as before, just starting from this new value of x (note a remains unchanged).

The sequence can end in two ways

1. A function gives $\frac{1}{2}$
2. The desired value of n is reached (if there is no desired n , some functions can run infinitely long)

Here, let $n = 5$. The process will be done for $a = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}$

Before starting, observe something about $ax(1-x)$. The maximum value of this function is when $x = \frac{1}{2}$ and decreases as x trends towards 0 or 1. So in cases where $f\left(\frac{1}{2}\right) < \frac{1}{2}$, all V will be 0.

For $a = \frac{1}{2}$:

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} \left(\frac{1}{2} \left(1 - \frac{1}{2}\right)\right) = \frac{1}{8} \implies V_1 = 0$$

All other V_i will be 0 since the maximum value is less than $\frac{1}{2}$

For $a = \frac{3}{2}$:

$$f\left(\frac{1}{2}, \frac{3}{2}\right) = \frac{3}{2} \left(\frac{1}{2} \left(1 - \frac{1}{2}\right)\right) = \frac{3}{8} \implies V_1 = 0$$

All other V_i will be 0 since the maximum value is less than $\frac{1}{2}$

For $a = \frac{5}{2}$:

$$f\left(\frac{1}{2}, \frac{5}{2}\right) = \frac{5}{2} \left(\frac{1}{2} \left(1 - \frac{1}{2}\right)\right) = \frac{5}{8} \implies V_1 = 1$$

$$f\left(\frac{5}{8}, \frac{5}{2}\right) = \frac{5}{2} \left(\frac{5}{8} \left(1 - \frac{5}{8}\right)\right) = \frac{75}{128} \implies V_2 = 1$$

$$f\left(\frac{75}{128}, \frac{5}{2}\right) = \frac{5}{2} \left(\frac{75}{128} \left(1 - \frac{75}{128}\right)\right) \approx 0.606 \implies V_3 = 1$$

All other V_i will be 1 since $a = \frac{5}{2}$ produces an attracting fixed point.

For $a = \frac{7}{2}$:

$$f\left(\frac{1}{2}, \frac{7}{2}\right) = \frac{7}{2} \left(\frac{1}{2} \left(1 - \frac{1}{2}\right)\right) = \frac{7}{8} \implies V_1 = 1$$

$$f\left(\frac{7}{8}, \frac{7}{2}\right) = \frac{7}{2} \left(\frac{7}{8} \left(1 - \frac{7}{8}\right)\right) = \frac{49}{128} \implies V_2 = 0$$

$$f\left(\frac{49}{128}, \frac{7}{2}\right) = \frac{7}{2} \left(\frac{49}{128} \left(1 - \frac{49}{128}\right)\right) \approx 0.827 \implies V_3 = 1$$

$$f\left(0.827, \frac{7}{2}\right) = \frac{7}{2} (0.827 (1 - 0.827)) \approx \frac{1}{2} (\text{slightly above}) \implies V_4 = 1$$

The above was just above $\frac{1}{2}$, so the cycle will more or less repeat, which implies $V_5 = 0$

2.6 Discussion on section 2.5

A few patterns observed for functions of order 1:

1. If the output of the map is always in the upper half or lower half of the function, it can be said that the result will be all 1s or 0s, respectively.
2. The values seem to trend towards attracting fixed points, if they exist. Note that the previous report defined fixed points to be attracting on $f(x)$ when $1 < a < 3$. This is noticeable in particular with $a = \frac{5}{2}$. For this a , there is an attracting fixed point at $x = \frac{3}{5}$, so values trended towards $\frac{3}{5}$. Because the attracting fixed point has an output above $\frac{1}{2}$, the terms of V will eventually be all 1s. If it had an output below $\frac{1}{2}$, then the terms of V will eventually be all 0s.
3. If the point repels, then the behavior seems to be a lot more erratic, consistently jumping between 1 and 0. But stronger repelling points seem to have more 0 V terms than 1 V terms after doing some experimentation.

3 Weeks 8-9

3.1 Premise

This section builds off of finding V terms by writing a script to computationally find them. It also explores order of sequences.

3.2 Python program to find first n terms of V on $f(x) = ax(1 - x)$

Below is Python code that takes in a value for a (such that $0 \leq a < 4$) and a value of n . The program outputs the first n terms for V as done in section 4 of the previous report.

```
import numpy as np

try:
    aInput = float(input("Define a value of a (such that 0 < a < 4): "))
    if aInput <= 0 or aInput > 4:
        raise Exception("Value must be such that 0 < a < 4")
    iterationsInput = int(input("Set the number of iterations: "))
except ValueError:
    print("Please enter a number.")
    quit()

def f(a,x):
    return a*x*(1-x)

xInput = 0.5
output = np.array([])

for i in range(0, iterationsInput):
    xInput = f(aInput, xInput)
    # print(f"xInput now: {xInput}") # (statement used in debugging)
    if xInput < 0.5:
        output = np.append(output, 0)
    else:
        output = np.append(output, 1)

print(output)
```

Sample run:

```
Define a value of a (such that 0 < a <= 4): 3.5
Set the number of iterations: 10
[1. 0. 1. 1. 1. 0. 1. 1. 1. 0.]
```

3.3 Parity-Lexicographical Ordering

Definition 3.1: Parity-lexicographical ordering is a comparison between two strings of the same length that are comprised of 0s and 1s. Traverse through each string until there is a mismatch (one string is 1 and the other has 0 for the n th character). If the number of 1s that precede this mismatch is even, then $0 < 1$ at this position. Otherwise, $0 > 1$ at this position.

Example 1: Let $v = 10111101$, $w = 10110101$. Determine which is greater.

Solution: Observe that the first four digits of the sequence are the same - 1011. Digit 5 varies, with $v_5 = 1$, $w_5 = 0$. Preceding digit 5 (so the sequence 1011), there are 3 1's, which is odd. Therefore, at digit 5, $0 > 1$, and since $w_5 = 0$ and $v_5 = 1$, $w > v$

Below is a python program that determines which of two strings is greater:

```
validInput = False
v = ""
w = ""
while not validInput:
    v = input("Give a string of only 0s and 1s: ")
    for i in range(0, len(v)):
        if v[i] != "0" and v[i] != "1":
            print("Invalid input.")
            break
    if i == len(v) - 1:
        validInput = True

validInput = False

while not validInput:
    w = input("Give another string of only 0s and 1s that is the same length as the first string: ")
```

```

    if len(v) == len(w):
        for i in range(0, len(w)):
            if w[i] != "0" and w[i] != "1":
                print("Invalid input.")
                break
            if i == len(w) - 1:
                validInput = True
    else:
        print("Invalid input.")

if v == w: # strings are the exact same
    print("The strings are equal.")
    quit()

numOnes = 0
mismatchVal = -1

# find the index where the strings differ
# without loss of generality (wlog), this runs until hitting the end of v
for i in range(0, len(v)):
    if v[i] != w[i]:
        mismatchVal = i
        break

for i in range(0, mismatchVal): # right before the index of the mismatch value
    if v[i] == "1": # wlog choose v (w will yield the same result)
        numOnes += 1

if numOnes % 2 == 0:
    if v[mismatchVal] == "1":
        print("The first string is greater")
        quit()
    else:
        print("The second string is greater")
        quit()
else: # odd number of ones
    if v[mismatchVal] == "0":
        print("The first string is greater")
        quit()
    else:
        print("The second string is greater")
        quit()

```

3.4 Python program to find values of $ax(1-x)$ given string of 0s and 1s

Progress so far (never completed it):

```

# this program takes in a string of 0s and 1s and comes to a conclusion on which
# types of functions can be mapped
# this first version assumes that the initial x is always 1/2
import numpy as np

validInput = False
v = ""
while not validInput:
    v = input("Give a string of only 0s and 1s: ")
    for i in range(0, len(v)):
        if v[i] != "0" and v[i] != "1":
            print("Invalid input.")
            break
        if i == len(v) - 1:
            validInput = True

def f(a,x):
    return a*x*(1-x)

if v[0] == "0":
    for i in range(1, len(v)):
        if v[i] == "1":
            print("No values exist")
            quit()
    print("Possible a values: 0 < a < 2")
    quit()

```

```

minA = 2
maxA = 4
# going down this route assumes that the string starts with 1
# idea: binary search algo
for i in range(1, len(v)):
    if v[i] == "1":
        print("") # some logic to reduce the range of A

```

4 Week 10

4.1 Premise

This section explores pre-images of functions. It also finds all possible values of a , if any, given a sequence of 0s and 1s of length 3. Finding values of a for a finite V length helps explore how this idea can be applied to a V of greater length.

4.2 Python program to find pre-image

Definition 4.1: The pre-image of a value x is any x^* in which taking the image of it yields x

Below is a Python program that prompts the user for an a and an x value, then finding all pre-image values of that x on the function $f(a, x) = ax(1 - x)$.

```

# this code takes an a value (0 < a <= 4) and an x value (0 <= x <= 1)
# given these parameters, the code calculates all pre-images (will be some x)
# when the image of these x values are taken, the starting value is obtained

from numpy.polynomial import Polynomial
xInput = 0
try:
    aInput = float(input("Define a value of a (such that 0 < a <= 4): "))
    if aInput <= 0 or aInput > 4:
        raise Exception("Value must be such that 0 < a <= 4")
    xInput = float(input("Define a value of x (such that 0 < x <= 1): "))
    if xInput <= 0 or xInput > 1:
        raise Exception("Value must be such that 0 < x <= 1")
except ValueError:
    print("Please enter a valid number.")
    quit()

# Polynomial solves for zeroes. The function ax(1-x) is converted into
# -ax^2 + ax - (inputted x)
# roots go from lowest to highest (x^0, x^1, x^2)

poly = Polynomial([-xInput, aInput, -aInput])
print(poly.roots())

```

4.3 All possible a for first three terms of V

Given the first 3 values of V , what can be said about the bounds of a ? This section explores that.

When $V = 000$:

If the first term is 0, since the starting point is at the maximum for the function, the image is still less than $\frac{1}{2}$. So subsequent values will also have images less than $\frac{1}{2}$, implying that if the first term is 0, subsequent values must also be 0. So if the maximum value of the function is less than $\frac{1}{2}$, then the following holds:

$$V = 000 \implies 0 < a < 2$$

When $V = 001, 010$, or 011 :

No values exist. As explained above, if the first value of 0, all subsequent values must also be 0. But these three terms start with 0 and have a 1 somewhere, which contradicts this rule.

When $V = 111$:

If there is a chain of 1s, this implies that the maximum value is above $\frac{1}{2}$, which occurs when $a > 2$. Also, this implies an attracting fixed point where the image of it is above $\frac{1}{2}$. Such fixed points occur when $2 < a < 3$ (remember from a prior report that attracting fixed points are defined from $1 < a < 3$).

But the chain of 1s continue to some point beyond $a = 3$. The fixed points are repelling at this point, but the repulsion isn't strong enough to cause the image of $ax(1 - x)$ to ever dip below $\frac{1}{2}$. An arbitrary a does exist where any values beyond it will have $ax(1 - x)$ dip below $\frac{1}{2}$ on the first image (so set $x = \frac{1}{2}$). Then:

$$\begin{aligned} ax(1 - x) &= \frac{1}{2} \\ ax^2 - ax + \frac{1}{2} &= 0 \end{aligned}$$

Next, use the quadratic formula with $a = 1, b = -1, c = \frac{1}{2}$. Discarding the negative solution because it doesn't make sense in this context, $a = 1 + \sqrt{5}$. Any value beyond this will lead to there being some 0s in the output. So,

$$V = 111 \implies 2 < a < 1 + \sqrt{5}$$

When $V = 101$:

Right after $a = 1 + \sqrt{5}$, the first values of V alternate between 0s and 1s (i.e. 101010...). As a increases, so do the frequency of 0s, but using mathematical brute force, the first three terms of V stay as 101 until $a \approx 3.8318 - 3.8319$ - at $a = 3.8318$, the first three terms are 101, but by $a = 3.1819$, the first three terms are 100. So,

$$V = 101 \implies 1 + \sqrt{5} < a < 3.1818 - 3.1819$$

When $V = 100$:

Once $V = 100$ at $a = 3.8318 - 3.8319$, it stays this way for the rest of the defined range of a since the third image value gets progressively lower. So,

$$V = 100 \implies 3.8318 - 3.8319 < a \leq 4$$

When $V = 110$:

No values exist. As previously discussed, at $a = 1 + \sqrt{5}$, the first three terms go from $V = 111$ to $V = 101$. At no a value is there a case where the first two (or more) terms of V are 1, only later being followed by a 0.

5 Week 11

5.1 Premise

This section expands on the previous section by exploring pre-pre-images (and beyond) and finding all possible a for the first four terms of V . It also introduces a potential order 3 polynomial that also has two critical points within the same map and is dependent on just a single parameter (similar to the degree 2 polynomial that was explored in-depth in previous sections).

5.2 Python program to find pre-image of pre-image (and beyond)

This program has two versions.

Version 1: This program finds the two pre-images of a specific x . Then it finds the pre-image values for the two pre-image values found from the original x (so four pre-pre-image values.) Then it finds all pre-pre-pre-image values, all pre-pre-pre-pre-image values, and so forth. (It basically does a level order traversal, with all pre-images being computed before any pre-pre-images, all pre-pre-images being computed before any pre-pre-pre-images.

The number of pre-image values grows exponentially, so iterating more than 30 times can start to take a bit.

Below is the Python code for this program:

```
# this code takes an a value (0 < a < 4) and an x value (0 < x < 1)
# given these parameters, the code calculates all pre-images (will be some x)
# when the image of these x values are taken, the starting value is obtained

# generally, a given value has two pre-images, though some will have no pre-images
# numpy will output complex numbers when there is no pre-image
# the program does not attempt to find the pre-image of an imaginary number
# at the critical point, there will be one pre-image - itself

# this code takes a level by level approach to find pre-images
# this means that all pre-pre images will be found before any pre-pre-pre images are found

import numpy as np
from numpy.polynomial import Polynomial

image_x_per_level = np.array([])

def pre_image(start_x):
    global image_x_per_level
    # Polynomial solves for zeroes. The function ax(1-x) is converted into
    # -ax^2 + ax - (inputted x)
    # roots go from lowest to highest (x^0, x^1, x^2)
    #print(f"Determining image of {start_x}")
    poly = Polynomial([-start_x, aInput, -aInput])
    if not np.iscomplex(poly.roots()[0]):
        # print(type(image_x_per_level))
```

```

        # print(type(poly.roots()))
    for root in poly.roots():
        # every pre-image value is put into image_x_per_level
        # these pre-image values are then processed on the next round
        image_x_per_level = np.append(image_x_per_level, root)
        # np.concatenate(image_x_per_level, poly.roots())
    print(f"Image of {start_x}: {poly.roots()}")
else:
    print(f"Image of {start_x} does not have any roots.")

def pre_image_manager(input_x):
    global image_x_per_level
    image_x_per_level = np.array([])
    for value in input_x:
        pre_image(value)

xInput = 0
try:
    aInput = float(input("Define a value of a (such that 0 < a ≤ 4): "))
    if aInput <= 0 or aInput > 4:
        raise Exception("Value must be such that 0 < a ≤ 4")
    xInput = float(input("Define a starting value of x (such that 0 < x ≤ 1): "))
    if xInput <= 0 or xInput > 1:
        raise Exception("Value must be such that 0 < x ≤ 1")
    # note: computation time really starts to grow above n = 30
    # takes ~1 min when printing is on, ~35 secs when printing is off
    iterations = int(input("Define number of iterations: "))
except ValueError:
    print("Please enter a valid number.")
    quit()

image_x_per_level = np.append(image_x_per_level, xInput)

for i in range(iterations):
    print(f"Level {i}: ".format(i+1))
    pre_image_manager(image_x_per_level)

Version 2: This program finds the two pre-images of a specific x, but it will only find the pre-pre-image based off the maximum root from that pre-image. This program takes a more depth-based traversal approach as opposed to the level-order traversal from the first version.

# this code takes an a value (0 < a ≤ 4) and an x value (0 ≤ x ≤ 1)
# given these parameters, the code calculates all pre-images (will be some x)
# when the image of these x values are taken, the starting value is obtained

# generally, a given value has two pre-images, though some will have no pre-images
# numpy will output complex numbers when there is no pre-image
# the program does not attempt to find the pre-image of an imaginary number
# at the critical point, there will be one pre-image - itself

# this code takes a level by level approach to find pre-images
# this means that all pre-pre images will be found before any pre-pre-pre images are found

import numpy as np
from numpy.polynomial import Polynomial

image_x_per_level = np.array([])

def pre_image(start_x, min_or_max):
    global image_x_per_level
    # Polynomial solves for zeroes. The function ax(1-x) is converted into
    # -ax^2 + ax - (inputted x)
    # roots go from lowest to highest (x^0, x^1, x^2)
    poly = Polynomial([-start_x, aInput, -aInput])
    if not np.iscomplex(poly.roots()[0]): # if one root is complex all roots will. complex roots
    # aren't valid
        if min_or_max == 0:
            return poly.roots()[0]
        else:
            return poly.roots().max()
    else:
        return -1

```

```

xInput = 0
mode = 0
try:
    aInput = float(input("Define a value of a (such that 0 < a ≤ 4): "))
    if aInput ≤ 0 or aInput > 4:
        raise Exception("Value must be such that 0 < a ≤ 4")
    xInput = float(input("Define a starting value of x (such that 0 ≤ x ≤ 1): "))
    if xInput < 0 or xInput > 1:
        raise Exception("Value must be such that 0 ≤ x ≤ 1")
    # note: computation time really starts to grow above n = 30
    # takes ~1 min when printing is on, ~35 secs when printing is off
    iterations = int(input("Define number of iterations: "))
    mode = int(input("Define mode of pre-image - 0 for min root, 1 for max root: "))
    if mode != 0 and mode != 1:
        raise Exception("Value must be 0 or 1")
except ValueError:
    print("Please enter a valid number.")
    quit()

image_x_per_level = np.append(image_x_per_level, xInput)

for i in range(iterations):
    print("Level {}: ".format(i+1))
    new_image = pre_image(xInput, mode)
    if xInput == -1:
        print(f"There are no pre-image values for {xInput}.")
        break
    if mode == 0:
        print(f"Min pre-image value of {xInput} is {new_image}")
    else:
        print(f"Max pre-image value of {xInput} is {new_image}")
    xInput = new_image

```

5.3 All possible a for first four terms of V

Given the first 4 values of V , what can be said about the bounds of a ? This section explores that.

When $V = 0000$:

If the first term is 0, since the starting point is at the maximum for the function, the image is still less than $\frac{1}{2}$. So subsequent values will also have images less than $\frac{1}{2}$, implying that if the first term is 0, subsequent values must also be 0. So if the maximum value of the function is less than $\frac{1}{2}$, then the following holds:

$$V = 0000 \implies 0 < a < 2$$

When V_1 is 0, but a later term is 1 (e.g. 0010)

No values exist. As explained above, if the first value of 0, all subsequent values must also be 0. But these three terms start with 0 and have a 1 somewhere, which contradicts this rule.

When $V = 1111$:

If there is a chain of 1s, this implies that the maximum value is above $\frac{1}{2}$, which occurs when $a > 2$. Also, this implies an attracting fixed point where the image of it is above $\frac{1}{2}$. Such fixed points occur when $2 < a < 3$ (remember from a prior report that attracting fixed points are defined from $1 < a < 3$).

But the chain of 1s continue to some point beyond $a = 3$. The fixed points are repelling at this point, but the repulsion isn't strong enough to cause the image of $ax(1-x)$ to ever dip below $\frac{1}{2}$. An arbitrary a does exist where any values beyond it will have $ax(1-x)$ dip below $\frac{1}{2}$ on the first image (so set $x = \frac{1}{2}$). Then:

$$\begin{aligned} ax(1-x) &= \frac{1}{2} \\ ax^2 - ax + \frac{1}{2} &= 0 \end{aligned}$$

Next, use the quadratic formula with $a = 1, b = -1, c = \frac{1}{2}$. Discarding the negative solution because it doesn't make sense in this context, $a = 1 + \sqrt{5}$. Any value beyond this will lead to there being some 0s in the output. So,

$$V = 1111 \implies 2 < a < 1 + \sqrt{5}$$

.

When $V = 1010$:

Right after $a = 1 + \sqrt{5}$, the first values of V alternate between 0s and 1s (i.e. 101010...). As a increases, there is a general increase in 0s overall, but the ordering of them isn't well defined, so the behavior from beyond the point of $a = 1 + \sqrt{5}$ is best known as chaotic. Mathematical brute force reveals that beyond

$V = 1010$, increasing the values of a will lead to $V = 1011$, then $V = 1010$, and finally $V = 1000$ up to $a = 4$.

The switchover from 1010 to 1011 occurs around 3.4985 - 3.4986. The switchover from 1011 to 1001 occurs around 3.8318 - 3.8319. Finally, the switchover from 1001 to 1000 occurs around 3.9602 - 3.9603.

When $V_1, V_2 = 1$ and a later V term is 0 (e.g. 1100) (3 terms):

No values exist. As previously discussed, at $a = 1 + \sqrt{5}$, the first three terms go from $V = 111$ to $V = 1010$. At no a value is there a case where the first two (or more) terms of V are 1, only later being followed by a 0.

5.4 Degree 3 polynomials with two critical points on map and single parameter

Objective: Find a degree 3 polynomial that has two critical points on the map

$$f : [0, 1] \rightarrow [0, 1]$$

and has the following fixed points:

$$f(0) = 0, f(1) = 1$$

and is dependent on only a and x .

First, to have $f(0) = 0$ and two critical points on the map, generally speaking, $f(1) = 1 - f(1) \neq 0$ - the function must be increasing when it enters the map (if it were decreasing at $f(0) = 0$, it wouldn't be in the map), and with two critical points, one will be decreasing and the other increasing for a degree 3 polynomial.

So for $f(x) = ax^3 + bx^2 + cx + d$, first note that $d = 0$ since the function passes through the origin. Next,

$$f(1) = a(1)^3 + b(1)^2 + c(1) = 1 \implies a + b + c = 1$$

Two critical points are chosen at $x = \frac{1}{3}$ and $x = \frac{2}{3}$ (though as will be seen, this doesn't seem to matter too much, as the critical points do change, possibly due to over-constraining the function. Regardless, there are 3 equations to work with and 3 unknowns, so write everything in terms of one of the variables (choosing a):

$$(1) f' \left(\frac{1}{3} \right) = 0 \implies \frac{1}{3}a + \frac{2}{3}b + c = 0$$

$$(2) f' \left(\frac{2}{3} \right) = 0 \implies \frac{4}{3}a + \frac{4}{3}b + c = 0$$

$$(3) f(1) = 1 \implies a + b + c = 1$$

By setting equations 1 and 2 equal to each other, cancel c out and obtain $b = -\frac{3}{2}a$. Plug the new term for b into equation 3 and solve for c , netting $c = 1 + \frac{1}{2}a$

Finally, plug this all into $f(x)$:

$$f(x) = ax^3 - \frac{3}{2}ax^2 + \left(1 + \frac{1}{2}a\right)x$$

By looking at the derivative, so

$$3ax^2 + 2\left(-\frac{3}{2}a\right)x + \left(1 + \frac{1}{2}a\right)$$

observe that these critical points form when the derivative is less than 0. This minimum is shown to always be at $x = \frac{1}{2}$ on a graphing calculator (leading theory is because this is the midpoint relative to the x values of the two critical point constraints).

So by plugging $\frac{1}{2}$ in for x on the derivative, setting the whole thing equal to 0, and solving for a , it's found that $a = 4$. Therefore, these critical points form when $a > 4$. Observing the function on a graphing calculator shows that the minimum and maximum for $f(x)$ on the defined map is when $a = 16$.

In conclusion,

$$f(x) = ax^3 - \frac{3}{2}ax^2 + \left(1 + \frac{1}{2}a\right)x$$

satisfies the initial objective when $4 \leq a \leq 16$.